Part 1

Math:

Define scene:

$c = (0.45m, 0.05, 0.15)\ m$ — sphere center

$|c| = \sqrt{0.45^2 + 0.05^2} = 0.477\ m$ — Base

$R = 0.10\ m$ — sphere radius

$a = (-0.3, -1, 1)$ — circle axis

$\beta = 35^\circ$ — circle angle

$L = 0.10m$ — Pen length

$N = 300$ — Number of Points

$0.477 > 0.10 \rightarrow$ Base is outside sphere (origin)

Coordinate System

$\hat{a} = \frac{a}{\|a\|} \rightarrow \|a\| = \sqrt{(-0.3^2) + (-1)^2 + (1)^2} = 1.4457$

$\hat{a} = \frac{(-0.3, -1, 1)}{1.4457} = (-0.2075, -0.6917, 0.6917)$ — PathNormalVector

first Plane: $k = (0,0,1)$

$\hat{a} \times k = (\hat{a}_y, -\hat{a}_x, 0)$

$= (-0.6917, 0.2075, 0)$

$|\hat{a} \times k| = \sqrt{(-0.6917)^2 + (0.2075)^2} = 0.7222$

$\hat{u} = \frac{(0.6917, 0.2075, 0)}{0.7222} = (-0.9578, 0.2873, 0)$ — UnitVector1

Surface Normal: Point on sphere

Surface Normal: $\rightarrow 0 \le \emptyset \le 2\pi$

$\hat{n}(\emptyset) = \cos\beta \cdot \hat{a} + \sin\beta(\cos\emptyset \cdot \hat{u} + \sin\emptyset \cdot \hat{v})$

w/ $30^\circ \rightarrow \cos\beta = 0.8660$

$\sin\beta = 0.5$

Point on Sphere:

PathCoords $o(\emptyset) = c + R \cdot \hat{n}(\emptyset)$

Circle center and radius:

$O_c = c + R\cos\beta \cdot \hat{a} = (0.4320, -0.0099, 0.2099)\ m$ — Circle center

$r = R\sin\beta = 0.10 \cdot 0.5 = 0.05\ (50\ mm)$ — Circle radius

second plane:

$\hat{v} = \hat{a} \times \hat{u}$

$V_x = -0.1988$

$V_y = -0.6625$

$V_z = -0.7222$

UnitVector2: $\hat{v} = (-0.1988, -0.6625, -0.7222)$

Check $\rightarrow$ $\hat{a} \times \hat{u} = 0$ ✓

$\hat{a} \times \hat{v} = 0$ ✓

$\hat{u} \times \hat{v} = 0$ ✓

Tangent Tool orientation

Tangent (direction of travel):

$\hat{t}(\emptyset) = -\sin\emptyset \cdot \hat{u} + \cos\emptyset \cdot \hat{v}$

Tool axes (spong(x_n, y_n, z_n)):

$z_{tool} = -\hat{n}(\emptyset)$

$x_{tool} = \hat{t}(\emptyset)$

$y_{tool} = z_{tool} \times x_{tool}$

Rotation Matrix:

$R(\emptyset) = [x_{tool}\ \ y_{tool}\ \ z_{tool}]$

$x_{tool} = \hat{t}(0) = \hat{v} = (-0.1988, -0.6625, -0.7222)$

$z_{tool} = -\hat{n}(0) = (0.6586, 0.4554, -0.5990)$

$y_{tool} = z \times x = (-0.7257, 0.5947, -0.3459)$

$\rightarrow$ Det(R) = 1 ✓

$\rightarrow R(0) = \begin{bmatrix} -0.1988 & -0.7257 & 0.6586 \\ -0.6625 & 0.5947 & 0.4554 \\ -0.7222 & -0.3459 & -0.5990 \end{bmatrix}$

$H^0_{tool}(0) = \begin{bmatrix} -0.1988 & -0.7257 & 0.6586 & 0.3841 \\ -0.6625 & 0.5947 & 0.4554 & 0.0045 \\ -0.7222 & -0.3459 & -0.5990 & 0.2099 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} x_{tool} & y_{tool} & z_{tool} \\ -0.1988 & -0.7257 & 0.6586 \\ -0.6625 & 0.5947 & 0.4554 \\ -0.7222 & -0.3459 & -0.5990 \end{bmatrix}$

Sampling the circle (constant speed)

$$\theta_k = \frac{2\pi(k-1)}{N}, \quad k = N$$

$$\Delta\theta = \frac{2\pi}{N} = \frac{2\pi}{300} = 0.0209 \text{ rad} \quad \boxed{1.2^\circ}$$

Arc length per step: $r \cdot \Delta\theta = 0.05 \cdot 0.0209 = 1.05 \text{ mm}$

Inverse kinematic

with Corke Toolbox in matlab